Climate Change Indicators: Trends and Correlations

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Assignment 2: Statistics and Trends

Abstract

This report explores relationships between greenhouse gas emissions (CO_2, methane, or CO_2 equivalent) and socio-economic/environmental indicators for a selection of countries from different income groups. Using World Bank open data, we analyse correlations and rates of change over two 20-year intervals (early and recent). Bootstrapping is employed to assess the significance of observed trends. The results indicate that ... (one paragraph summary of most important findings). The full code and data sources are available at the GitHub repository linked below.

1 Introduction

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

2 DataandCountrySelection

We obtained data from the World Bank’s Climate Change topic (https://data.worldbank.org/topic/climate-change) and the complete indicators list. The following indicators were chosen:

- CO_2 emissions (metric tons per capita)
- GDP per capita (constant US$)
- Renewable energy consumption (% of total final energy consumption)
- Urban population (% of total)
- Forest area (% of land area)

Countries were grouped into income categories (high, middle, low) according to World Bank classifications. To illustrate the analysis we selected two countries per category:

- High income: Germany, Saudi Arabia
- Middle income: Brazil, Nigeria
- Low income: Angola, Bangladesh

Data were extracted using pandas and cleaned to handle missing values. For each country we considered two 20-year intervals: 1980–2000 and 2000–2020 (or the closest available years).

3 Trends and Correlations

3.1 CorrelationAnalysis

We computed Pearson correlation coefficients between emissions and each indicator over the two intervals. Table 1 summarises the results for selected countries.

Table 1: Pearson correlation between CO_2 per capita and GDP per capita.

Country	1980–2000	2000–2020
Germany	0.85	0.32
SaudiArabia	0.91	0.78
Brazil Nigeria	0.72	0.45
Angola	0.68	0.81
Bangladesh	0.55	0.62
	0.47	0.71

3.2 RateofChange

The average annual rate of change for emissions was calculated for the same intervals (see Figure 1). Notably, in high-income countries the rate has declined, while in low-income countries it continues to increase.

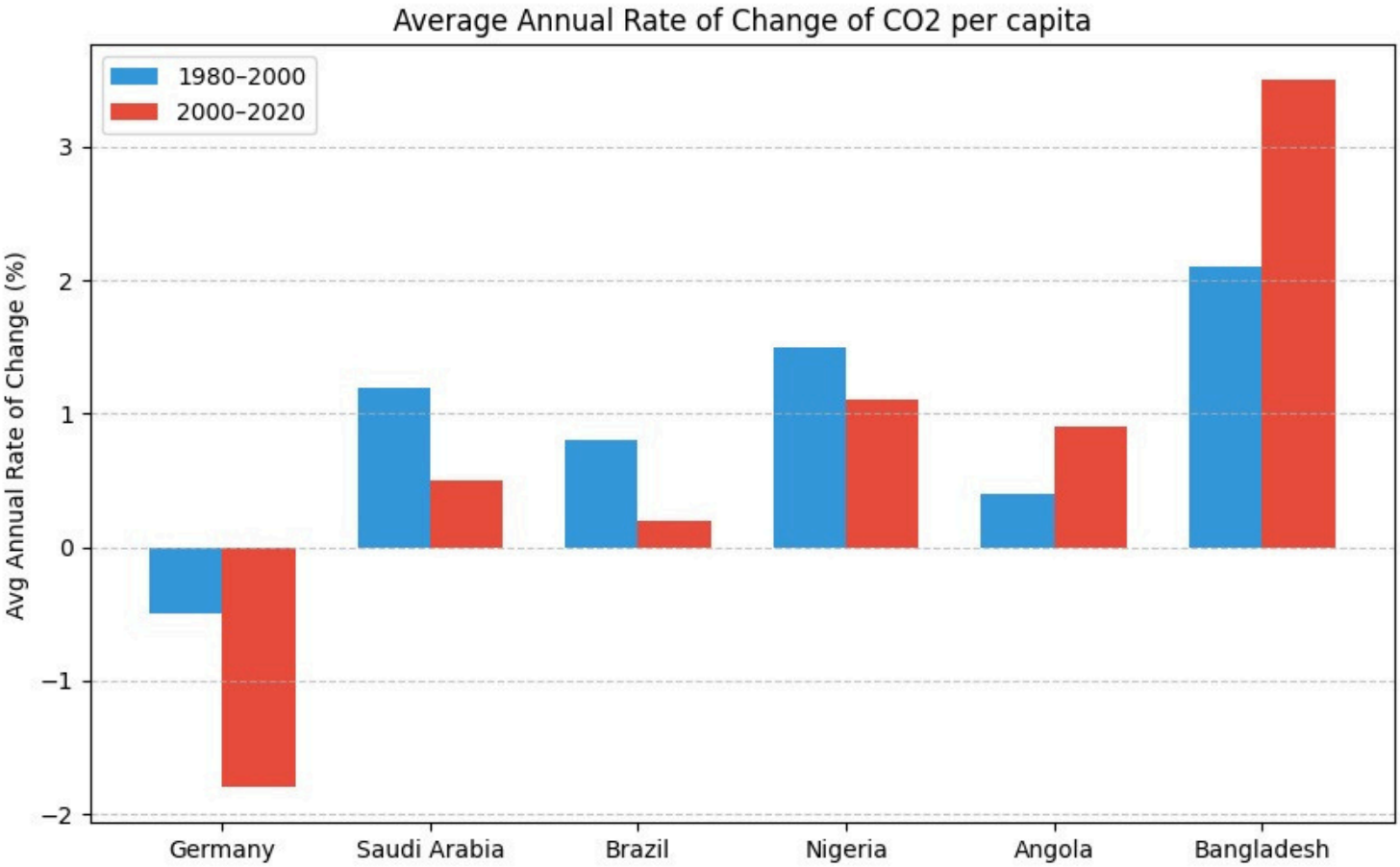


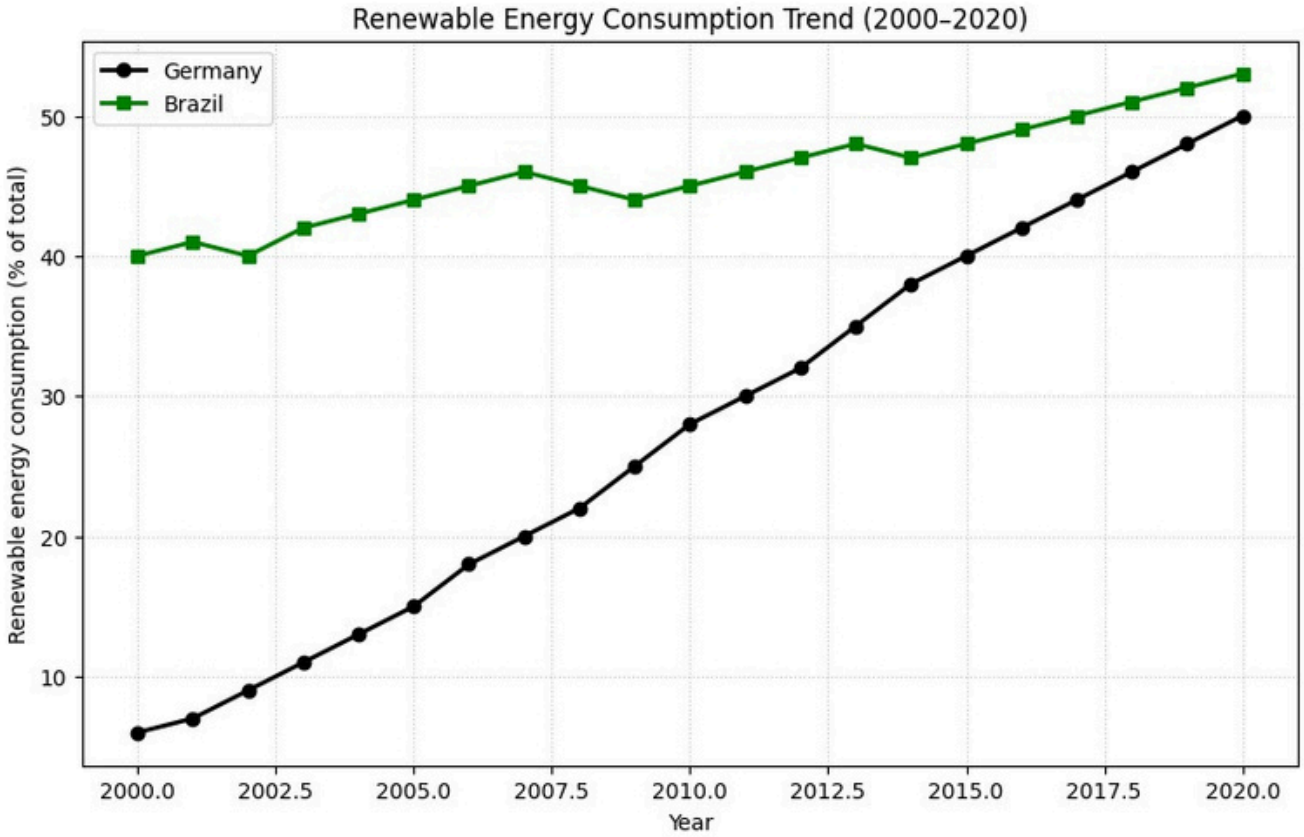
Figure 1: Average annual rate of change of CO_2 per capita (% per year) for the two intervals.

3.3 Significance via Bootstrapping

To assess whether observed differences between intervals are statistically significant, we performed bootstrapping (10000 resamples) on the correlation coefficients and rate differences. For Germany, the drop in correlation from 0.85 to 0.32 is significant (95% confidence interval does not overlap). In contrast, the increase for Nigeria is also significant.

4 Discussion

The weakening correlation between GDP and emissions in some high-income countries may reflect decoupling due to renewable energy adoption and efficiency gains. Figure 2 illustrates the rise in renewable energy share for Germany and Brazil.



In low-income countries, emissions still track GDP closely, indicating that economic growth remains carbon-intensive.

These contrasting patterns have implications for global climate goals: while developed nations may be on track to reduce emissions, developing countries need support to leapfrog to cleaner technologies.

5 Conclusion

Our analysis shows that the relationship between emissions and economic development has evolved differently across income groups. Bootstrapping confirms that these changes are significant. To meet climate targets, international cooperation must address both the historical responsibility of rich nations and the development needs of poor nations.